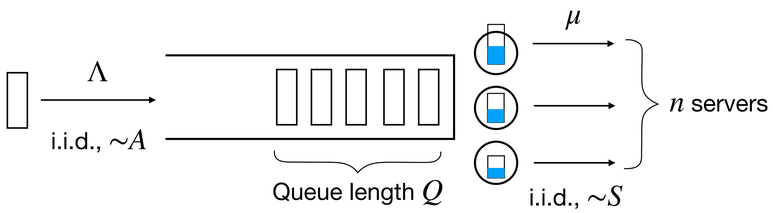
A new $1/(1-\rho)$-scaling bound for multiserver queues via a leave-one-out technique

Yige Hong, Computer Science Department, Carnegie Mellon University

Carnegie Mellon University

Model: GI/GI/n queue under FCFS



- Residual arrival time R_a ; residual service time R_i
- Load $\rho = \dfrac{\Lambda}{n\mu}$

How does the steady-state $\mathbb{E}[Q]$ scale when n is large?

Goal: show $\mathbb{E}[Q] \le O\left(1/(1-\rho)\right)$ for arbitrary $\rho < 1$ and n

Prior work

- Most prior work focuses on a particular scaling regime, e.g.

Classic heavy-traffic:	Halfin-Whitt:	Non-Degenerate Slowdown
$\rho \uparrow 1$, n fixed, $\mathbb{E}[Q] = O(1/(1-\rho))$	$\rho = 1-\Theta(n^{-1/2}), n\to\infty$ $\mathbb{E}[Q] = O(n^{1/2})$	$\rho = 1-\Theta(n^{-1}), n\to\infty$ $\mathbb{E}[Q] = O(n)$
[Kingman 62, 70] [Borokov 65] [Iglehart and Whitt 70] [Kollerstrom 74, 79] [Loulou 73] [Scheller-Wolf 03] [Hokstad 85] [Grosof et al. 22]…	[Hafin and Whitt 81] [Reed 09] [Gamarnik and Momcilovic 08] [Gamarnik and Stolyar 12] [Braverman and Dai 15, 16] [Aghajani and Ramanan 20]…	[Whitt 03] [Atar and Solomon 11] [Atar 12]…

$\mathbb{E}[Q] = O\left(1/(1-\rho)\right)$ for each scaling regime

- **Li and Goldberg' 25**: first **universal bound** with order $O\left(1/(1-\rho)\right)$, but with a 10^{21} constant

Our results

Previous universal bound	Our universal bound
When $\mathbb{E}[A^2] < \infty, \rho < 1, \mathbb{E}[S^{2+\epsilon}] < \infty$	When $\mathbb{E}[A^2] < \infty, \rho < 1, R_s^{\max} < \infty$
$\mathbb{E}[Q] \le \dfrac{2.1\times 10^{21}\times M}{1-\rho}$	$\mathbb{E}[Q] \le \dfrac{2R_s^{\max} + \mathrm{Var}(\Lambda A) - \rho}{2(1-\rho)} + \left(\mathbb{E}[(\Lambda A)^2]/2 - R_a^{\min}\right)$

where

- $M = \mathbb{E}[(\mu S)^2] \times \left(\mathbb{E}[(\mu S)^2]^{1+\epsilon} + \mathbb{E}[(\mu S)^{2+\epsilon}]\right) \times \left(\frac{1}{\epsilon}\right)^4 + 49\mathbb{E}[(\Lambda A)^2]$
- $R_s^{\max} = \sup_{t\ge 0} \mathbb{E}[\mu S - t \,|\, \mu S \ge t]$ $\quad R_a^{\min} = \inf_{t\ge 0} \mathbb{E}[\Lambda A - t \,|\, \Lambda A \ge t]$

In M/GI/n, our bound implies: $\mathbb{E}[Q] \le \dfrac{R_s^{\max}}{1-\rho}$

- $\mathbb{E}[Q] \le \dfrac{\max(1/\alpha, 1)}{1-\rho}$ if $S \sim$ Gamma with shape parameter α
- $\mathbb{E}[Q] \le \dfrac{\mu \max_{i\in[d]} \tau_i}{1-\rho}$ if $S \sim$ Phase-type; τ_i = expected complet time from phase i
- $\mathbb{E}[Q] \le \dfrac{1}{1-\rho}$ if $S \sim$ NBUE (e.g. Increasing Failure Rate, Erlang, Weibull with $k \ge 1$)
- $\mathbb{E}[Q] \le \dfrac{\mu S_{\max}}{1-\rho}$ if S is bounded by $S_{\max}$

Our results (heterogeneous)

When $\mathbb{E}[A^2] < \infty, \rho < 1, R_{s,i}^{\max} < \infty$ for all i

$$\mathbb{E}[Q] \le \frac{2\sum_{i=1}^n (\mu_i/\mu_\Sigma) R_{s,i}^{\max} + \mathrm{Var}(\Lambda A) - \rho}{2(1-\rho)} + \left(R_a^{\min} - \mathbb{E}[(\Lambda A)^2]/2\right)$$

where

- $\mu_\Sigma = \sum_{i=1}^n \mu_i$
- $R_{s,i}^{\max} = \sup_{t\ge 0} \mathbb{E}[\mu_i S_i - t \,|\, \mu_i S_i \ge t]$
- $R_a^{\min} = \inf_{t\ge 0} \mathbb{E}[\Lambda A - t \,|\, \Lambda A \ge t]$

Bounding Γ_1 in the challenging case

$$\Gamma_1 \triangleq \mathbb{E}_s\left[1\{Q=0\}R_1\right] - (1-\rho)\mathbb{E}\left[R_1\right]$$

$$= \mathbb{E}_s\Big[\underbrace{\left(1\{Q=0\} - 1\{Q_{-1}=0\}\right)R_1}_{\le R_s^{\max}/n}\Big] + \underbrace{\mathbb{P}_s\left[Q_{-1}=0\right]\mathbb{E}_s\left[R_1\right] - (1-\rho)\mathbb{E}\left[R_1\right]}_{= -\mathbb{E}[S^2]/(2n)}$$

$$= \left(R_s^{\max} - \mathbb{E}[S^2]/2\right)/n \quad \textbf{Q.E.D.}$$

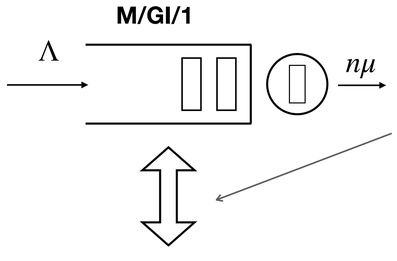
Proof (for M/GI/n)

Step 1: Comparing M/GI/n with M/GI/1

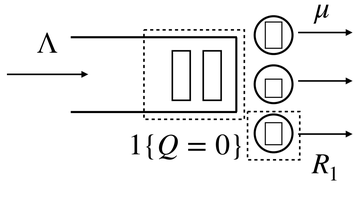
$$\mathbb{E}[Q] \le \underbrace{\frac{\mathbb{E}[S^2]}{2(1-\rho)}}_{\approx \mathbb{E}[Q]^{\mathrm{M/GI/1}}} + \frac{n}{1-\rho}\boxed{\Gamma_1}$$

$$\Gamma_1 = \mathbb{E}_s\left[1\{Q=0\}R_1\right] - (1-\rho)\mathbb{E}\left[R_1\right]$$

- $\mathbb{E}_s$: "Palm expectation" observed at completions
- $\Gamma_1 \approx$ **Covariance**$(1\{Q=0\}, R_1)$
- Want to show $\Gamma_1 = O(1/n)$



Modified M/GI/n
(completions = i.i.d. renewal processes)



- Intuitively, same arrivals and completion rates except when $Q = 0$
- Proved using the stationarity of the test function $\left(Q + \mu\sum_{i=1}^n R_i\right)^2$
- Formally, "Basic Adjoint Relationship"

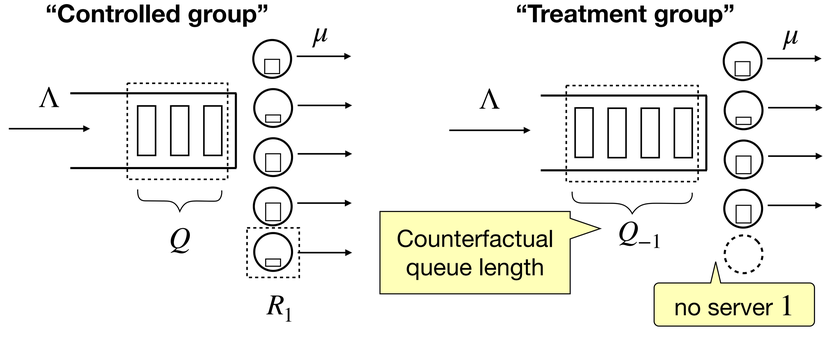
Finish the proof when $n \le 1/(1-\rho)$:

$$\Gamma_1 \le \mathbb{P}_s\left[Q=0\right]\mathbb{E}_s\left[R_1 \,|\, Q=0\right] - (1-\rho)\mathbb{E}\left[R_1\right]$$
$$\le (1-\rho)R_{s,\max} - (1-\rho)\mathbb{E}\left[R_1\right]$$
$$\le \frac{1}{n}R_{s,\max} - \frac{1}{2n}\mathbb{E}\left[S^2\right] \quad \textbf{Q.E.D.}$$

Step 2 (Key challenge): asymptotic independence

WTS "covariance" $\Gamma_1 = O(1/n)$ **when** $n > 1/(1-\rho)$

"Controlled group" **"Treatment group"**



$$\mathbb{E}_s\left[1\{Q=0\} - 1\{Q_{-1}=0\}\right] \le O(1/n) \quad \text{and} \quad Q_{-1} \perp R_1$$
$$1\{Q=0\} \ge 1\{Q_{-1}=0\} \text{ a.s.}$$